

32-bit microcontroller

FLASH operation instructions

Targets

Series	Product Model	Series	Product Model	Series	Product Model
HC32L110	HC32L110C6UA	HC32F00	HC32F003C4UA	HC32L13	HC32L130E8PA
	HC32L110C6PA		HC32F003C4PA		HC32L130F8UA
	HC32L110C4UA		HC32F005C6UA		HC32L130J8TA
	HC32L110C4PA		HC32F005C6PA		HC32L136J8TA
	HC32L110B6PA		HC32F005D6UA		HC32L136K8TA
	HC32L110B4PA				
	HC32L110B6YA				
HC32F03	HC32F030E8PA	HC32L07	HC32L072PATA	HC32F07	HC32F072PATA
	HC32F030F8UA		HC32L072KATA		HC32F072KATA
	HC32F030F8TA		HC32L072JATA		HC32F072JATA
	HC32F030H8TA		HC32L073PATA		
	HC32F030J8TA		HC32L073KATA		
	HC32F030K8TA		HC32L073JATA		
HC32L17	HC32L176PATA	HC32F17	HC32F176PATA	HC32L19	HC32L196PCTA
	HC32L176MATA		HC32F176MATA		HC32L196MCTA
	HC32L176KATA		HC32F176KATA		HC32L196KCTA
	HC32L176JATA		HC32F176JATA		HC32L196JCTA
	HC32L170JATA		HC32F176JATA		HC32L190JCTA
	HC32L170FAUA		HC32F170FAUA		HC32L190FCUA
HC32F19	HC32F196PCTA				
	HC32F196MCTA				
	HC32F196KCTA				
	HC32F196JCTA				
	HC32F190JCTA				
	HC32F190FCUA				

目 录

1	摘要.....	3
2	FLASH 简介	3
3	FLASH 应用说明	4
3.1	简介	4
3.2	安全特性	4
3.2.1	操作源保护	4
3.2.2	操作目标保护	4
3.2.3	PC 地址擦写保护	4
3.2.4	寄存器写保护	5
3.3	功能说明	5
3.4	工作流程介绍	6
3.4.1	Sector 擦除.....	6
3.4.2	Chip 擦除.....	6
3.4.3	写操作	7
3.4.4	读操作	7
3.5	基于 FLASH 安全特性的编程方法	8
3.5.1	基于 Keil MDK 的编程方法	8
3.5.2	基于 IAR 的编程方法.....	8
3.5.3	结果查看与示例.....	8
4	总结.....	9
5	版本信息 & 联系方式	10

1 Summary

This application note mainly introduces the FLASH security features, operation instructions and application methods of Xiaohua Semiconductor MCU*.

Note:

- *This application note is a supplementary application material for Xiaohua Semiconductor MCU* and cannot replace the user manual. Please refer to the user manual for specific functions and register operations and other related matters.*

2 FLASH Introduction

What is FLASH?

A type of flash memory device, flash memory is a non-volatile memory that can retain data for a long time even without current supply. Its storage characteristics are equivalent to hard disks. This characteristic is the basis for flash memory to become the storage medium of various portable digital devices.

(Quoted from 'Baidu Encyclopedia', 'Interactive Encyclopedia', 'Wikipedia')

What are the characteristics of FLASH?

Flash memory is a non-volatile memory that can erase and reprogram memory unit blocks called blocks. The write operation of any Flash device can only be performed in empty or erased units,

Application of FLASH?

FLASH is widely used in emerging digital devices such as mobile storage, MP3 players, digital cameras, and handheld computers.

***See cover for supported models.**

3 FLASH Application Description

3.1 Introduction

Xiaohua Semiconductor MCU* covers 16/32/64/128/256/512K Byte FLASH memory according to different models. Each FLASH is divided into sectors, and each sector has a capacity of 512 bytes. This memory supports erase (chip/page erase), programming, and read operations. In addition, this module also supports erase and write protection of FLASH memory, as well as write protection of control registers.

3.2 Security Features

3.2.1 Operation Source Protection

Xiaohua Semiconductor MCU* adopts high-security hardware design for FLASH with a capacity of more than 32K, and has FLASH operation source defense function: FLASH erase and write operations can only be performed correctly when the address of the FLASH operation function is located in 0~32K. FLASH address 0~32K has higher security, and important functions must be placed in this area, such as important program entry, interrupt entry function, high-security algorithm module, UID, AES, true random number, RTC algorithm coordination, to form a high-security authentication system.

3.2.2 Operation Target Protection

The entire 64K byte FLASH memory is divided into 128 pages, and every 4 pages share an erase protection bit. When a page is protected, the erase and write operations on the page are invalid, and an alarm flag and an interrupt signal are generated. When any page in the FLASH memory is protected, the full chip erase and write of the FLASH is invalid, and an alarm flag and an interrupt signal are generated.

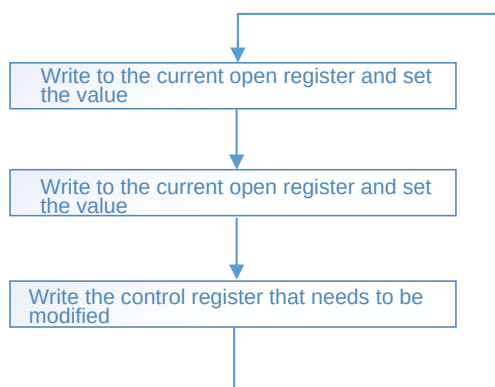
3.2.3 PC Address Erase and Write Protection

When the CPU runs a program in FLASH, if the current PC pointer happens to fall within the page address range to be erased, the erase and write operation is invalid and an alarm flag and an interrupt signal are generated.

**See cover for supported models.*

3.2.4 Register Write Protection

The controller of this module shields ordinary write operations and must be modified by writing sequence. The specific operation steps are shown in the figure below:



Note:

- Write 0x5a5a, 0xa5a5, write the target register. No write operation (write ROM, RAM, REG) can be inserted between these three write operations, otherwise the value of the target register cannot be rewritten. If the rewrite fails, you need to repeat these three steps.

3.3 Functional Description

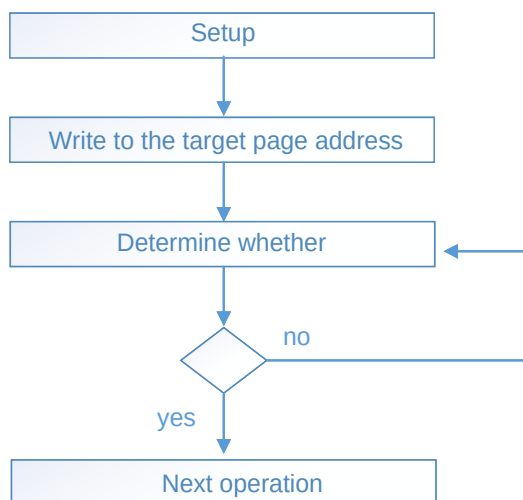
This section introduces the functions, workflow and programming methods based on security features of the FLASH controller module.

This FLASH controller supports three bit width read and write operations of eFLASH: Byte (8 bits), Half-word (16 bits), and Word (32 bits). Note that the address of Byte operation must be aligned to Byte, the target address of Half-word operation must be aligned to Half-word (the lowest address bit is 1'b0), and the address of Word operation must be aligned to Word (the lowest two addresses are 2'b00). If the address of the read or write operation is not aligned according to the bit width specification, the operation is invalid and the system will enter a Hard Fault error interrupt.

3.4 Workflow Introduction

详细操作步骤，请参考相应系列的用户手册。

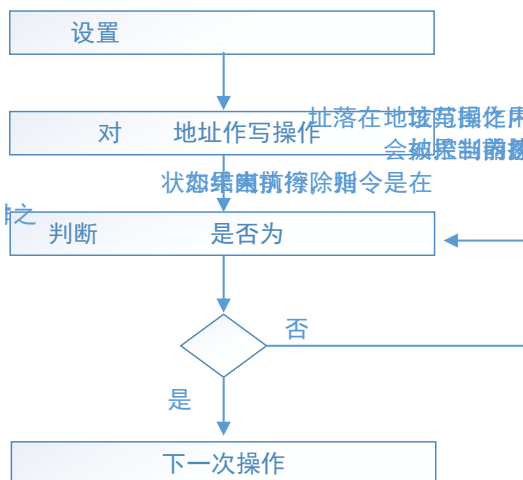
3.4.1 Sector Erase



Note:

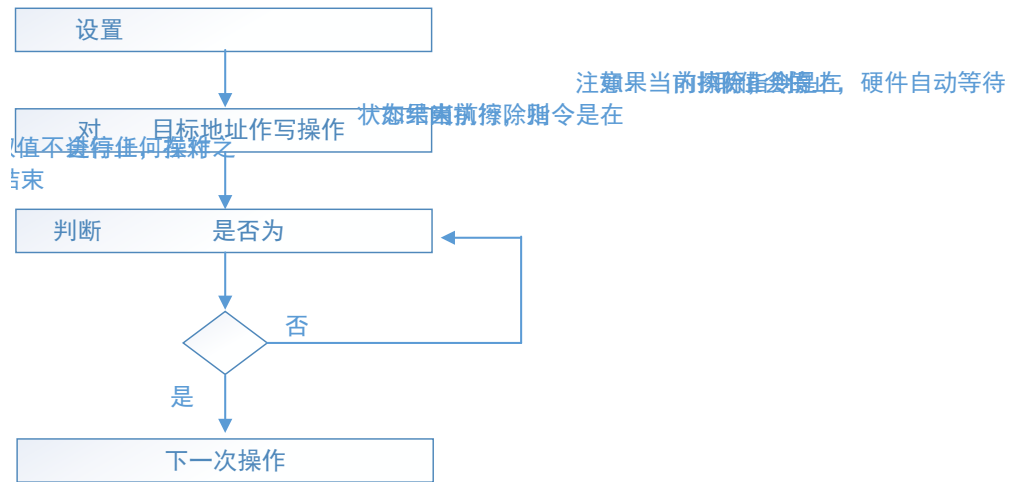
The controller ignores the lower bits of the target address as long as the target address falls within the address range of the page. This write operation is used to trigger the page erase operation, and the written data will also be ignored by the controller. If the current erase instruction is executed in , the value will stop, and the hardware will automatically wait for the state of to end. If the current erase instruction is executed in , the value will not stop. Before performing any operation on , the software must determine whether the state of has ended.

3.4.2 Chip Erase

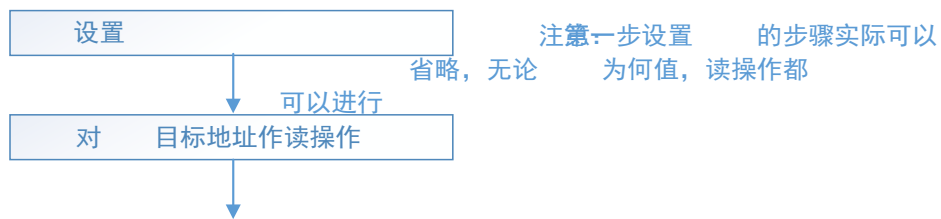


注意：控制器忽略目标地址的低地址落在该范围作用即触发页擦除操作，所写数据也会被控制器忽略。在硬件自动等待状态结束前，擦除指令是在

3.4.3 Write Operation



3.4.4 Read Operation



3.5 Programming method based on FLASH security features

在实际应用中，对于容量大于 32K 的 MCU，如果需要将 FLASH 的操作函数、安全功能函数等放置在 FLASH 的 32K 安全区，可通过以下便捷的方式实现。

说明：

- 本例为说明需要，主要示例将函数“Flash_SectorErase()”放置在安全区“0x400”地址的方法，实际应用当中可以将“示例函数”和“地址”根据自己的需求进行替换。

3.5.1 Programming method based on Keil MDK

在 Keil MDK 中，可以简单通过如下方式实现对安全函数的执行地址映射：

在目标函数的声明处增加以下代码：

```
en_result_t Flash_SectorErase(uint32_t u32SectorAddr) __attribute__((section(".ARM._at_0x400")));
```

3.5.2 Programming method based on IAR

1、在目标函数定义处增加以下代码：

```
en_result_t Flash_SectorErase(uint32_t u32SectorAddr) @".Flash_SectorErase"
```

2、在工程“*.icf”文件中增加以下代码：

```
place at address mem:0x00000400 { readonly section .Flash_SectorErase};
```

3.5.3 Result viewing and examples

如果对此方法产生的具体结果信息感兴趣，可以通过 map 文件、调试文件或在调试运行期间观测代码的执行地址空间。

示例如下（可以确认目标函数确实放在了预期[0x400]地址）：

Functions					flash.c	
Name	Address	Size	#Insts	Source		
*	*	*	*	*		
Flash_Init	0000 0180	156	71	flash.c:229		
Flash_LockAll	0000 0214	48	24	flash.c:627		
FLASH_RAM_IRQHandler	0000 0174	12	6	interrupts_b		
Flash_SectorErase	0000 0400	144	64	flash.c:499		
Flash_UnlockAll	0000 0238	48	22	flash.c:647		
Flash_WriteByte	0000 0260	144	70	flash.c:287		
HardFault_Handler	0000 02E4	16	8	interrupts_b		
HardFault_Handler	0000 00F0	2	1	sysctrl.c:19		
I2C0_IRQHandler	0000 02F4	8	4	interrupts_b		
I2C1_IRQHandler	0000 02FC	8	4	interrupts_b		
LCD_IRQHandler	0000 0304	8	4	interrupts_b		
LPTIM0_1_IRQHandler	0000 030C	12	6	interrupts_b		
LPUART0_IRQHandler	0000 0318	8	4	interrupts_b		

496	** \retval ErrorInvalidParameter	FLASH地址无效
497	** \retval ErrorTimeout	操作超时
498	*****	
499	en_result_t Flash_SectorErase(uint32_t u32SectorAddr)	
500	{	
	0000 0400 PUSH (R3-R7, LR)	
	0000 0402 MOV R4, R0	
501	en_result_t	enResult = Ok;
502	volatile uint32_t	u32TimeOut = FLASH_TIMEOUT_ERASE;
	0000 0404 MOVS R0, #0xFF	
	0000 0406 STR R0, [SP]	
	0000 0408 MOVS R0, #4	
503		

4 Summary

以上章节主要介绍了小华半导体 MCU*的 FLASH 的安全特性、工作流程和基于 FLASH 安全特性的编程方法，在实际开发中用户可以根据以上方法进行修改或扩展来满足自己的应用。

*支持型号见封面。

5 Version information & contact information

日期	版本	修改记录
2019/8/19	Rev1.0	初版发布
2022/7/15	Rev1.1	公司 Logo 更新。



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